



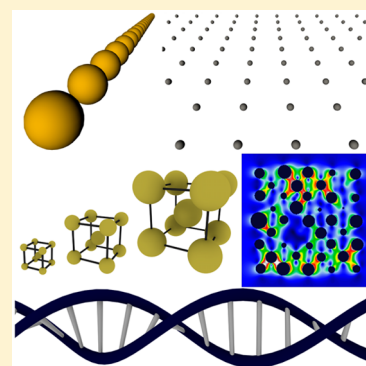
Optical Properties of One-, Two-, and Three-Dimensional Arrays of Plasmonic Nanostructures

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S Supporting Information

ABSTRACT: This Feature Article describes research on the optical properties of arrays of silver and gold nanoparticles, particles that exhibit localized surface plasmon resonances in the visible and near-infrared. These resonances lead to strong absorption and scattering of light that is strongly dependent on nanoparticle size and shape. When arranged into multidimensional arrays, the nanoparticles strongly interact such that the collective properties can be rationally designed by changing the dimensions of the array (one-, two-, or three-dimensional), interparticle spacing, and array shape or morphology. Emerging from this work is a large body of literature focusing on one-, two-, and three-dimensional arrays, which provide unique opportunities for realizing materials with interesting and unusual photonic and metamaterial properties. Electrodynamics theory provides an accurate description of the optical properties, often based on simple models such as coupled dipoles, effective medium theory, and anomalous diffraction. In turn, simple models and simulation methods allow for the prediction and explanation of a variety of observed optical properties. In one and two dimensions, these tunable optical properties range from extinction spectra that are red- or blue-shifted compared to the isolated particles to lattice plasmon modes that involve strong interactions between localized plasmon resonances in the nanoparticles and photonic modes that derive from Bragg diffraction in the crystalline array. Three-dimensional arrays can exhibit unique effective medium properties, such as negative permittivity that leads to metallic optical response even when there is less than 1% metal content in the array. They also can be rationally designed to have photonic scattering modes dictated and controlled by interactions between nanoscale plasmonic nanoparticles and the mesoscale superlattice crystal habit (i.e., the crystalline size, shape, and morphology). This discussion of plasmonic arrays across multiple dimensions provides a comprehensive description of those factors that can be easily tuned for the design of plasmon-based optical materials.



I. INTRODUCTION

The development of nanostructured materials has spurred tremendous interest in the rational design of structures with new and unusual properties. Unlike bulk materials, when the size, shape, or surface chemistry of a nanomaterial is changed, it can dramatically alter its properties. For example, bulk metals are glossy and reflective, hence their use in coinage and jewelry; however, on the nanoscale many metals exhibit the ability to confine and guide light, resulting in materials with beautiful and vibrant colors. Gold and silver nanoparticles were used as early as the fourth century by the Romans to color structures like the Lycurgus Cup (unbeknownst to them), which contains metallic nanoparticles and transmits vibrant ruby red but reflects emerald green light.¹ Metal nanoparticles were also commonly used in Medieval times to create the stained glass windows in cathedrals.² The striking appearance of metal nanoparticles stems from the remarkably efficient interaction between such particles and incident light, termed the localized surface plasmon resonance (LSPR).³ In addition, the optical properties of metal nanoparticles are highly dependent on their proximity to other nanoparticles and as such can be rationally designed by controlling their environment, arrangement, and symmetry.

Thus, a grand challenge in the field of plasmonics has been to develop techniques for controllably assembling metal nanostructures in two and three dimensions as well as methods for understanding and ultimately designing the properties of such hierarchical materials *a priori*.

In 1908, Gustav Mie presented the analytic solution to Maxwell's equations for a small sphere.⁴ Mie's solution, coupled with modern computing power, means that the rapid and exact solutions to extinction, scattering, and the local-electric field of a sphere are trivial to obtain.^{5,6} However, understanding light-matter interactions for metal nanoparticles that are of arbitrary shape, in very close proximity, or in large three-dimensional structures that are structurally complex is challenging both computationally and conceptually.^{3,7} Thus, the development of accurate theories and simple heuristic models is crucial to understanding and designing plasmonic arrays with novel optical properties. These theories have been central to realizing a variety of applications involving noble metal nanoparticles,

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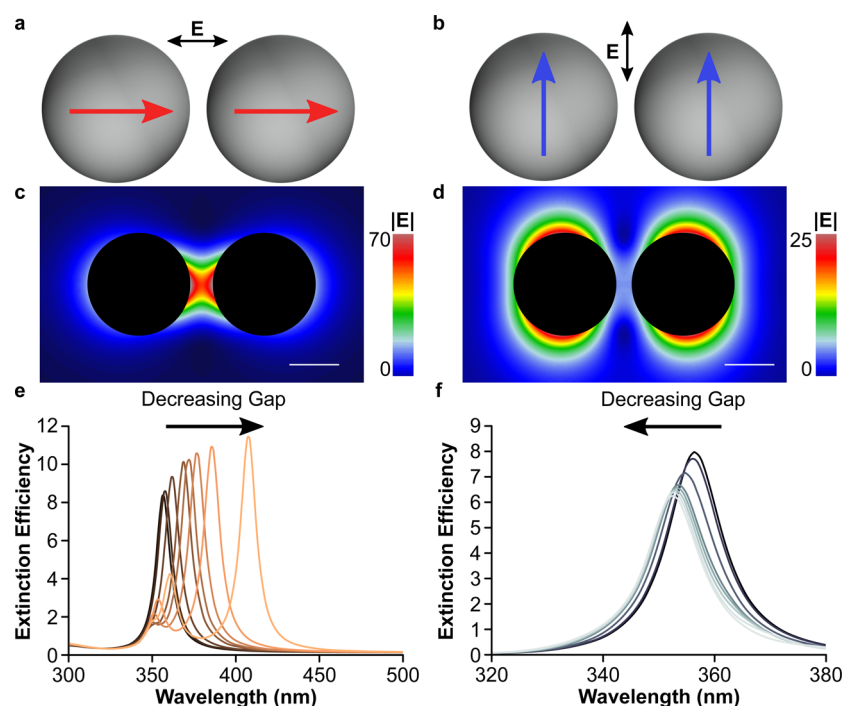


Figure 1. Plasmon coupling in spherical dimers with small gaps. (a) Constructive dipolar coupling in spheres with light polarized along the dimer axis. (b) Destructive interference between dipoles with light polarized across the dimer axis. Electric field intensity of constructively (c) and destructively (d) interfering 10 nm radius Ag spheres separated by a 5 nm gap. Extinction spectra of a 10 nm radius sphere Ag dimer with decreasing interparticle gaps with light polarized along (e) and across (f) the dimer axis. The interparticle gaps in (e) and (f) (from darkest to lightest trace) are 50, 40, 30, 20, 10, 5, 4, 3, 2, and 1 nm. The scale bar is 10 nm.

especially in the areas of molecular sensing,^{8–10} metamaterials,^{11–13} spectroscopic enhancement,^{14–18} and light harvesting.^{19–21}

In this Feature Article, we focus on the optical properties of one- (1D), two- (2D), and three- (3D) dimensional plasmonic nanoparticle arrays and incorporate insights from recent experimental and theoretical work. A consistent theme throughout these discussions is that many of the properties of these arrays can be understood solely based upon simple dipolar models. We begin by briefly giving an overview of the LSPR resonance condition and its properties. We also consider dimers of particles, and then we develop the theory for 1D and 2D arrays. Thereafter, we survey studies that have explored plasmonic nanoparticles on 2D surfaces, including work on photonic lattice modes, plasmon-laser cavities, and unusual plasmonic retardation effects. Finally, we explore plasmonic nanoparticle arrays in three dimensions, primarily focusing on those assembled using DNA, an area in which we have played a central role. Here, we survey the physical models and methods with which one can predict and understand the structure and function of such assemblies, illustrate a few of their properties, and investigate the interplay between the nanoscale plasmonic building blocks and shape-derived photonic effects. We anticipate that this Feature Article will be a useful reference and roadmap for understanding and advancing the development of novel plasmonic arrays.

II. LOCALIZED SURFACE PLASMON RESONANCE

To begin, we present an overview of the localized surface plasmon resonance (LSPR) in metal nanoparticles. In a small metallic particle, excitation by incident light at the proper frequency drives a collective oscillation of the conduction electrons. The wavelength at which the LSPR occurs can vary

considerably, depending on the composition, size, shape, and dielectric environment of the single particle; we will address each of these factors briefly.³

For a small sphere with radius a , the extinction efficiency for the dipole resonance is

$$Q_{\text{ext}} = \frac{24\pi a}{\lambda} \frac{\epsilon_i \epsilon_m}{\epsilon_i^2 + (\epsilon_r + 2\epsilon_m)^2} \quad (1)$$

where ϵ_r and ϵ_i are the real and imaginary parts of the dielectric function, respectively; λ is the wavelength; and ϵ_m is the background dielectric constant.^{3,5} Several noteworthy characteristics of the LSPR can be inferred from this expression. First, the resonance condition is achieved when $\epsilon_r = -2\epsilon_m$; negative ϵ_r is a requirement for supporting a surface plasmon resonance. Metals such as Au, Ag, Cu, Pd, and Al are the most common plasmonic metals due to their relatively high quality resonances;^{3,22–25} however, a variety of other materials with sufficient carrier density also can support LSPRs such as indium tin oxide (ITO)^{26,27} and certain doped semiconductors.^{28,29} On resonance, the strength of the LSPR is limited by the magnitude of ϵ_i , which is indicative of loss in the material. The wavelength at which the LSPR occurs can be shifted by changing the size, background medium, or shape of the particle. Increasing the background index ϵ_m changes the wavelength at which $\epsilon_r = -2\epsilon_m$ occurs, resulting in a red-shift of the LSPR.

For particles with radii similar to $\lambda/2\pi$, the quasistatic expression above does not fully describe retardation effects due to the large sphere size.^{3,5} Various theories, including the Modified-Long-Wavelength Approximation (MLWA), have been developed to describe spheres of this size in a simple manner.^{3,30–32} The MLWA allows one to capture retardation effects for spheres or spheroids, including radiative damping,

dynamic depolarization, and higher-order multipoles, all of which are a result of the electric field oscillating across the particle volume. These effects primarily red-shift and broaden the resonance. Increased scattering by the particle is the primary cause of broadening because photons that are re-emitted (i.e., scattered) by the particle compete with radiation that excites the LSPR, quenching the resonance.

The effect of shape on the LSPR is more complex. Geometric effects significantly alter the near-field interaction of light with anisotropic nanoparticles. For example sharp tips promote high intensity electric fields and strong far-field scattering that significantly alter the position and intensity of the resonance.^{3,33,34} Visualizing the interaction of the incident field with the nanoparticle is often necessary to gain a physical understanding of the LSPR in addition to explaining surface-enhanced processes such as SERS^{35,36} and fluorescence quenching.^{9,37} For further information on the LSPR in anisotropic nanoparticles, we direct the reader to several important reviews.^{3,38,39}

III. ONE- AND TWO-DIMENSIONAL ARRAYS OF PLASMONIC NANOPARTICLES: THEORY

In this section, we consider the theoretical description of nanoparticle dimers as well as coupled linear chains and 2D arrays of spherical particles. Dimers are of interest when there are small gaps (1–10 nm) between nanoparticles, as this can lead to enhanced near fields that are important in a variety of photophysical processes.⁴⁰ In addition, the behavior of dimers can be understood by modeling dipolar interactions, providing a useful conceptual framework that can then be generalized to one-, two-, and three-dimensional arrays.

1. Plasmonic Dimers. The dimer of spheres has long served as an ideal structure for the development of new theories to understand plasmon coupling^{41–43} and for studying spectroscopic enhancement and spectral shifts.^{35,40,44} In particular, there can be red or blue shifts in plasmon resonance wavelengths as the particles approach as a result of dipolar coupling that is very much analogous to shifts that occur between aggregated organic dyes. Kasha et al. identified that there were two types of dye aggregates that can form in solution, so-called J-aggregates and H-aggregates.^{45,46} In different solvent configurations the dyes can assemble in either head-to-tail or face-to-face orientations, resulting in either red (J-) or blue (H-) shifted absorption. This insight regarding dye photophysics has changed our understanding of excitonic interactions in polymeric semiconductor materials,⁴⁷ and it also provides inspiration for understanding LSPR behavior as well. This complements other models and frameworks that have been developed for coupled plasmonic nanoparticles to explain spectral shifts and changes in lifetimes⁴⁸ including energy orbital diagrams,⁴⁹ molecular orbital formalism,^{41,50} and quantum coupling methods,^{40,51–53} to name a few.

Here we describe plasmonic interactions using a simple dipole–dipole model, which is easily generalized to understand multidimensional arrays. If the incident field is polarized along the axis of the dimer where the dipoles are additive (Figure 1a), a strongly confined, intense electromagnetic field arises in the gap (Figure 1c). Constructive interference between the additive dipoles results in red-shifting of the LSPR that increases with smaller gaps (Figure 1e). Conversely, if the light is polarized transverse to the spheres (Figure 1b), the dipoles destructively interfere, and a less intense electric field is observed in the gap (Figure 1d). Here, the LSPR blue-shifts and decreases in

intensity (Figure 1f). Note that when the particles are separated by more than $\lambda/2\pi$ the static dipolar interactions are modified by retardation effects, leading to changes and even reversals of these trends. This phenomenon is explained in detail below.

2. One-Dimensional and Two-Dimensional Diffractive Plasmonic Arrays. The optical properties of nanoparticle arrays, linear or otherwise, exhibit richer properties than isolated dimers. Two general classes of chain arrays can be considered: (1) those coupled in the near-field and (2) those that permit far-field photonic modes to interact with the plasmonic particles. The first class leads to red shifts and blue shifts in spectra that are similar to those described above for the dimer case, and this can lead to strong fields at gaps between particles due to localized resonances.^{54–59} We will focus primarily on the second class, as this leads to new modes known as “lattice plasmons” that have hybrid photonic/plasmonic character. Notable lattice plasmon behavior includes remarkably narrow line widths and enhanced local electric fields. One important feature of lattice plasmons is that they are delocalized over a large number of particles.

Lattice plasmon modes occur when the geometric arrangement and periodicity of an array allow for multiple diffractive scattering events to occur between particles.^{60–63} Many of the concepts regarding dipolar array periodicity and far-field extinction were first presented by Markel using an analytic theory that works in 1D and by Carron in an application involving surface-enhanced Raman scattering for 2D nanoparticle arrays with varying pitch.^{60,62} In an array, the nanoparticle LSPR provides a strong oscillator and source of pronounced scattering while the periodicity determines which scattered wavelength will lead to constructive interference. The effect of periodicity is similar to what occurs in a photonic crystal; however, the nanoparticle polarizability is much larger for plasmonic oscillators, which leads to strong coupling effects that are distinct from arrays of dielectric particles. To describe lattice plasmons, we will briefly survey the coupled dipole method, which provides a framework through which physical insights regarding the far-field optical properties can be understood.^{16,60,63–66} Subsequently, in this section and the next, we survey experimental and theoretical observations in this growing field.

In the coupled dipole approximation, an array of N identical particles is considered where $\mathbf{r}_i$ and α_s represent the position and polarizability of a given dipole, respectively. The induced dipole $\mathbf{P}_i$ in each particle in the presence of an applied plane wave field is $\mathbf{P}_i = \alpha_s \mathbf{E}_{\text{loc},i}$ (where $i = 1, 2, \dots, N$), where the local field $\mathbf{E}_{\text{loc},i}$ is the sum of the incident and retarded fields of the other $N - 1$ dipoles. At any given wavelength λ the field is equal to

$$\mathbf{E}_{\text{loc},i} = \mathbf{E}_{\text{inc},i} + \mathbf{E}_{\text{dipole},i} \quad (2)$$

$$\mathbf{E}_{\text{loc},i} = \mathbf{E}_0 \exp(i\mathbf{k} \cdot \mathbf{r}_i) - \sum_{\substack{j=1 \\ j \neq i}}^N \mathbf{A}_{ij} \cdot \mathbf{p}_j, \quad i = 1, 2, \dots, N \quad (3)$$

where E_0 and $k = 2\pi/\lambda$ are the amplitude and wave vector of the incident wave, respectively. The interaction matrix between dipoles $\mathbf{A}$ is expressed as

$$\mathbf{A}_{ij} \cdot \mathbf{P}_j = k^2 e^{ikr_{ij}} \frac{\mathbf{r}_{ij} \times (\mathbf{r}_{ij} \times \mathbf{P}_j)}{r_{ij}^3} + e^{ikr_{ij}} \frac{(1 - ikr_{ij})[\mathbf{r}_{ij}^2 \mathbf{P}_j - 3\mathbf{r}_{ij}(\mathbf{r}_{ij} \cdot \mathbf{P}_j)]}{r_{ij}^5}, \quad i, j = 1, 2, \dots, N$$

$$, \quad j \neq i \quad (4)$$

where $\mathbf{r}_{ij}$ is the vector between dipole i and dipole j . Note that in eq 4 the first interaction term has $1/r$ dependence and describes radiative dipolar coupling between particles,⁶³ while the second has $1/r^2$ and $1/r^3$ dependence and evolves into the static dipolar coupling in the long-wavelength and near-field limits. In the case of a 1D infinite array, a simple analytic solution to the dipolar interaction matrix $\mathbf{A}$ can be derived that allows the derivation of the effective polarization induced in each particle

$$P = \frac{E_0}{\alpha_s^{-1} - S} \quad (5)$$

and the extinction cross-section for each particle

$$C_{\text{ext}} = 4\pi k \operatorname{Im}\left(\frac{P}{E_0}\right) = 4\pi k \operatorname{Im}\left(\frac{1}{\alpha_s^{-1} - S}\right) \quad (6)$$

where S is the retarded dipole sum that is related to the geometric arrangement of the particles

$$S = \sum_{j \neq i} \left[\frac{(1 - ikr_{ij})(3 \cos^2 \theta_{ij})e^{ikr_{ij}}}{r_{ij}^3} + \frac{k^2 (\sin^2 \theta_{ij})e^{ikr_{ij}}}{r_{ij}} \right] \quad (7)$$

and θ_{ij} is the angle between the polarization vector (in the plane of the array) and $\mathbf{r}_{ij}$ is the vector from dipole i to dipole j . Equations 2–4 apply to 1D, 2D, and 3D; however, only for the 1D case has an analytical evaluation been presented.⁶⁰ However, the numerical evaluation of eq 7 can be done for truncated arrays.

From eqs 5 and 6, we learn that the retarded dipole sum S greatly influences the optical properties of the array. Zou and Schatz⁶⁷ demonstrated that the real part of S has a logarithmic singularity when the wavelength λ matches the dipole spacing D and that this leads to peaks and dips in the extinction spectra, with dips found at wavelengths to the blue of the isolated particle plasmon wavelength and (generally) peaks when it is to the red. In this case the peaks arise when the real part of the denominator in eq 6, i.e., $\operatorname{Re}(\alpha_s^{-1} - S)$, vanishes. Another factor that contributes is $\operatorname{Im}(S)$, which becomes negative over a range of wavelengths $\lambda > D$ (Figure 2a, inset). This makes the difference $\operatorname{Im}(\alpha_s^{-1} - S)$ smaller, which reduces the line width (Figure 2a). Physically this behavior arises because light scattering from each particle can be coherently rescattered an infinite number of times for frequencies that satisfy the Bragg condition, resulting in confinement of the light for long times. This results in remarkably sharp and dispersive lineshapes that have interesting fundamental and technological implications.

Using the coupled dipole approximation, a variety of interesting optical phenomena have been identified that can be controlled with the nanoparticle size, spacing, and interparticle distance. As mentioned above, the most striking spectral feature of lattice plasmon modes is the remarkably narrow line width (Figure 2a) of the resonant peak, routinely ~ 1 meV and possibly narrower (in theory).^{63,65,66} As such, 2D

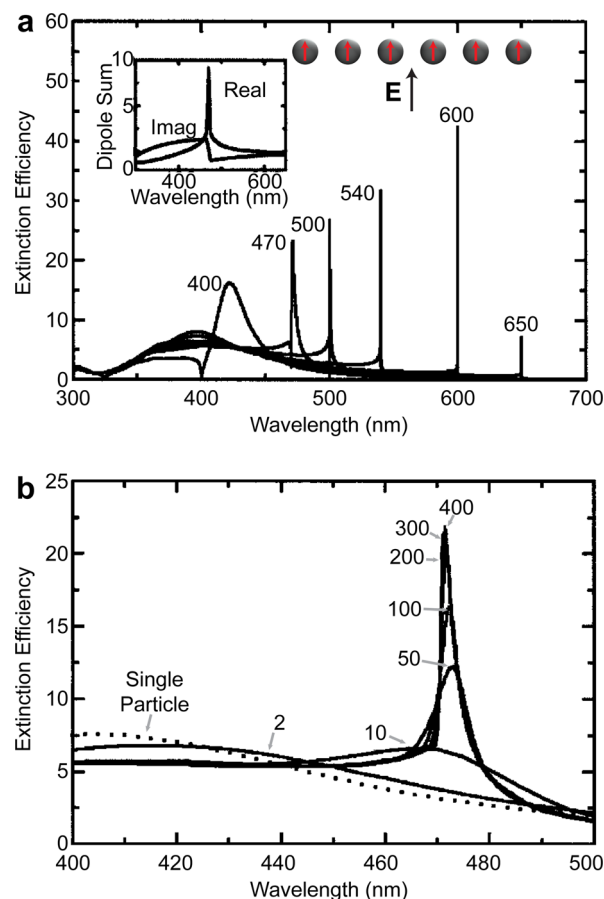


Figure 2. Narrow line widths in one-dimensional plasmonic arrays. (a) Calculated extinction of infinite linear chains of 50 nm radius Ag nanospheres with varying interparticle spacings. The inset depicts the real and imaginary dipole sums S for the 470 nm spacing. (b) Calculated extinction of finite linear chains of 50 nm radius Ag nanospheres with 470 nm spacing where the particle number varies from 1 to 400. All spectra were calculated using the coupled dipole approximation where the dipoles are described by the Mie polarizability for a sphere. Reprinted in part with permission from ref 63 (Zou, S.; Janel, N.; Schatz, G. C., Silver Nanoparticle Array Structures That Produce Remarkably Narrow Plasmon Lineshapes. *J. Chem. Phys.* **2004**, *120*, 10871–10875). Copyright 2004, American Institute of Physics.

arrays can provide enhanced sensing or detection opportunities due to enhanced sensitivity of the lattice plasmon to changes in the dielectric environment of the particle and local fields that are further amplified compared to the plasmon enhancement in the isolated nanoparticles.^{16,65} Because detection sensitivity is dependent on both the peak width and the extent of peak shift, narrower peaks provide lower detection limits even when the magnitude of the overall peak shifts is smaller.⁶⁵ Past work has estimated that the index of refraction sensitivity is a factor of up to nine times larger for lattice plasmons,⁶⁵ and there are also larger SERS enhancement factors¹⁶ (by factors of 10 – 10^3 depending on the array structure).

3. Structural Parameters That Control Lattice Plasmon Modes. Here, we outline the major physical features that influence the optical response of diffractive plasmonic arrays. The first requirement is that the periodicity (spacing) of the particles is commensurate with the wavelengths (or the wavelength/index when there is an elevated medium around the array) associated with the single-particle LSPR. For

example, if the LSPR band of the free nanoparticle spans 400–500 nm, the periodicity of the array would need to be ~ 400 –500 nm to observe a lattice plasmon mode. Figure 2a plots simulated lattice plasmon modes for one-dimensional arrays of 50 nm radius Ag spheres with spacings ranging from 400 to 650 nm (the isolated particle spectrum can be seen in Figure 2b, dashed line). Each of the lattice plasmon modes has spacings within the single LSPR “window”. Notably, the lattice plasmon mode with the highest extinction and narrowest peak occurs at 600 nm, red of the individual LSPR maximum. This is due to coherent mixing of the LSPR and photonic modes, where the latter is a Bragg-like diffractive mode defined by the periodicity of the lattice.^{65,68} The most intense lattice plasmon involves a compromise between maximizing the particle polarizability, which occurs at the LSPR wavelength, and minimizing the width, which occurs far to the red of the LSPR where the cancellation of radiative damping by the period array is most effective.

In addition to the periodicity of the array, both the particle size and array dimension must be sufficiently large to observe strongly dispersive extinction spectra. Both of these sufficient conditions derive from the need for the dipole sum (the sum of coherent scattering events by each particle) to be large enough to zero out the real part of the denominator in eq 5. Past work has shown that these diffractive lattice modes actually exist for all particle sizes; however, they are subtle (often too small to measure) for Ag spheres that are smaller than approximately 30 nm in radius.^{63,67,69} The primary effect of changing particle size is on the polarizability α_s , where larger particles are more polarizable and exhibit a stronger optical response. Regarding lattice size, a requisite number of particles is required to have sufficient coherent scattering events to produce a large enough real part and negative enough imaginary part of the dipole sum S (Figure 2b). For 50 nm radius spheres and 470 nm spacing, it was shown that as few as 50 particles in an array produce a narrow lattice plasmon band (10 nm width) that decreases in breadth as the array length trends toward infinity.⁶³ Importantly, both the effects of particle size and lattice dimension are easily understood through the interplay between the geometric dipolar lattice sum S and the polarizability of a plasmonic sphere α_s .

With a single nanoparticle LSPR, the width of the resonance is controlled by many factors including shape, crystallinity, radius of curvature, material composition, and many other factors.^{3,70} However, the rational arrangement of plasmonic nanoparticles provides a clear means for spectral control that cannot be achieved at the single nanoparticle level.

4. Broadening Effects in Two-Dimensional Plasmonic Arrays. One major distinction between the optical properties of one-dimensional and two-dimensional nanoparticle arrays is that the two-dimensional arrays consistently exhibit broader extinction profiles than one-dimensional chains. Zou and Schatz observed this effect for both cubic and hexagonal lattices of Ag spheres with a variety of spacings.⁶⁵ This effect can be understood qualitatively via a dipolar model. In 1D, the dipoles are strictly additive; adding more particles increases the constructive interference, sharpening the resonance. In 2D, however, there is a mix of constructive and destructive interference between dipole rows, reducing the overall dipole sum. This leads to the observed broadening and dampening of the collective resonances.

IV. ONE- AND TWO-DIMENSIONAL PLASMONIC ARRAYS: EXPERIMENTAL STUDIES

Experimentally, many high-throughput methods for synthesizing 1D and 2D arrays over a wide area have been developed. In this section, we describe attempts to observe lattice plasmons in these structures, as well as recent extensions of this work to plasmon lasers.

1. Initial Experimental Attempts at Observing Lattice Plasmon Modes. Many of the initial attempts to experimentally observe and control diffractive lattice plasmon modes had limited success; indeed, the experimental results did not produce the remarkably sharp resonances described above. Hicks et al. explored the collective resonances of 1D arrays of cylindrical Ag nanoparticles with diameters of 100 nm or greater.⁷¹ In the experiment, the arrays were immersed in index-matched oil to produce a uniform dielectric environment and excited by polarized white light. The white light is Rayleigh scattered by the arrays and collected through a dark-field condenser. Much of this initial difficulty in observing the predicted narrow line widths was attributed to differences between the experimental and simulation configurations. In particular, the spread in the excitation and collection angles in the experiment reduced coupling between particles, only exciting a subset of the array (Figure 3a). This effect was

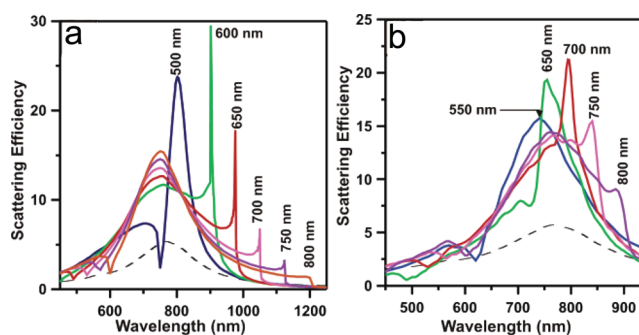


Figure 3. Early observations of line width control in one-dimensional plasmonic arrays. (a) Simulations of a 500 particle linear chain of Ag cylinders (height = 30 nm, d = 130 nm) at 0° incidence and (b) simulations of a 20 particle linear chain of Ag cylinders (height = 30 nm, d = 130 nm) with 30° incidence collection angle. The dashed line is the scattering spectra for the single particle times 2. The experimental data more closely resembled (b) as a result of the collection parameters used. Reprinted in part with permission from ref 71. Copyright 2005, American Chemical Society.

corroborated using DDA periodic boundary simulations,⁷¹ demonstrating that the greatly diminished array size reduces coherent coupling between particles, decreasing the lattice plasmon mode strength (Figure 3b). Disorder, polydispersity in the nanoparticle size, and “stitching” or lattice mismatch from the electron-beam lithography were shown to broaden the lattice plasmon modes considerably. Previous theoretical work had also shown that up to 10% disorder in the particle placement has only a minimal effect on the lattice plasmon mode, but larger amounts of disorder broaden and diminish the intensity of the resonance.⁶⁵

In work that somewhat pre-dated the Hicks et al. measurements, Haynes et al. identified that two-dimensional arrays of cylinders with large interparticle spacings exhibit unique blue-shifted resonances and narrowing.⁷² However, unlike the lattice plasmon mode, these effects were due to size-

dependent interference effects associated with roughly half-wavelength separations between the particles where the e^{ikr} terms in eq 4 can lead to sign reversals (when $kr \sim \pi$) in the usual dipolar interactions. This work was an important early experimental exploration of far-field plasmonic–photonic interactions; Supporting Information Section S.1 explores the implications and theory of these data in depth. These early works, coupled with more recent advances in lithographic and synthetic techniques, have enabled the realization of lattice plasmon modes with sharp diffractive peaks in 2D arrays.

2. Experimental Observations of Lattice Plasmon Modes. Recent experiments exhibit strong agreement with the initial coupled dipole predictions and corroborate coupled dipole and electrodynamics simulations.^{68,73–76} Results from Barnes and co-workers provide excellent agreement using an analytic ellipsoid polarizability that incorporates retardation effects to describe nanorod systems;⁶⁸ the work of Crozier uses large gold disks and verifies a sharp linear extinction response that changes with interparticle spacing;⁷³ and the Grigorenko data explore the role of Wood's Anomaly, the angular dependence on phase shifts in the diffractive modes, and the more recently demonstrated high-quality factor resonances in different solvents.⁷⁷ In all cases, the use of a homogeneous dielectric environment, proper excitation and collection angles, and appropriate choice of particle size and aspect ratio to maximize the interparticle diffraction events produced diffraction effects that proved challenging to see in early works.^{71,72}

Recent work by Zhao and Odom has demonstrated another effect in lattice plasmon arrays.⁷⁸ Rather than a single large diffractive peak, a subradiant resonance effect is observed due to interference between the sharp “in-plane” lattice plasmon modes and the broad “out-of-plane” modes.⁷⁸ These two different modes can be attributed to the anisotropy of the cylindrical particles used: it is well-known that anisotropic nanoparticles have longitudinal and transverse resonances.^{3,5,49,79}

As noted earlier, the lattice plasmon modes involve multiple scattering events between particles in the array, which enables an overall higher quality factor resonance.^{63,80} Effectively, the “recycling” of photons through successive scattering events provides a mechanism to recover some of the inherent loss in metal LSPRs, a concern in the use of plasmonics for spectral enhancement.^{19,81} In addition, this makes lattice plasmon arrays ideal for stimulated emission applications, such as was recently demonstrated by Zhao et al. in the development of 2D plasmon-enhanced lasers.⁸⁰ Plasmon lasers (*a.k.a.* spasers^{82,83}) are of interest due to their nanoscale geometries as they push the fundamental spatial limits of laser technology. In a plasmon laser, the near field at the nanoparticle surface stimulates the emission of light at a wavelength that approximately matches the lattice plasmon. This creates a new mode in which the dye exciton, the plasmon, and the photonic lattice mode all interact. There is also plasmon enhancement of the laser dye excitation, but this occurs at a shorter wavelength that is removed from the lattice plasmon and is therefore a less important effect. As proof that lattice plasmons play an important role in these lasers, the experiments (and theoretical modeling) were repeated for lattices composed of the nonplasmonic scatterers Ti or TiO₂, and no lasing emission was observed (Figure 4). It was also demonstrated that for Ag and Au nanoparticles stimulated emission produces a beam of light perpendicular to the surface, with narrow angular divergence.^{63,65,80,84} Thus, lattice plasmons

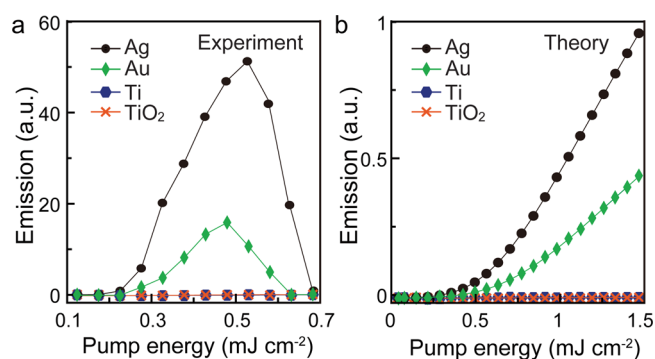


Figure 4. Stimulated emission from two-dimensional lattices of metallic and dielectric particle arrays. (a) Measured emission and (b) simulated emission of a two-state dye system of lattices composed of Ag, Au, Ti, and TiO₂. Differences at higher pump energies between experiment and theory are likely due to off-angle amplified spontaneous emission that competes with stimulated emission and from surface-normal lasing emission that is suppressed. Lasing was achieved at 913 nm (full-width at half-maximum 1.3 nm) dye using IR-140 dye and pulsed 800 nm laser light. Data adapted from ref 80.

show promise not only as a means of sensing but also more generally for the enhancement of photophysical processes including Raman scattering,^{16,44} lasing,⁸⁰ and fluorescence,³⁷ via plasmonically enhanced near fields.

Lattice plasmons are not limited to silver and gold nanoparticles. Recently, Olson et al. demonstrated the use of arrays of Al nanorods fabricated by electron-beam lithography to create “plasmonic pixel” arrays.⁸⁵ Using diffractive lattice plasmon modes to narrow the far-field scattering of 2D plasmonic arrays, it was shown that the colors achievable exceed those of current light-emitting diode (LED) technology. Additionally, arrays of Al pyramids have been used to control the directional emission of dye molecules at their surface, such that the emission intensity could be controlled based on orientation of the array.⁸⁶ It is likely that Al will play a significant role in future plasmonic applications due to its wide spectral activity, large and unique radiative scattering cross sections, and abundance and low cost.^{24,25,86–89}

V. THREE-DIMENSIONAL PLASMONIC ARRAY STRUCTURES

Three-dimensional lattices of plasmonic nanoparticles provide a means for introducing a vast and complex design space for constructing novel optical materials. There are many different ways to make such arrays, most of which are restricted to closest-packed structures where the particle diameter and spacing are directly connected. Here we focus on programmable assembly methods based upon oligonucleotide-linked nanoparticle structures;^{90–108} these methods provide a platform where the nanoparticle size, spacing, and crystal symmetry can all be controlled independently. In Supporting Information Section S.2, we provide a discussion describing other 3D array structures.

The initial work in this area was inspired by the remarkable color change observed in 1996 by Mirkin et al., when it was shown that DNA could be used to deliberately and programmably (by virtue of sequence) assemble gold nanoparticles into periodic structures.^{90,91,106} The free DNA-modified particles suspended in solution were the typical pink associated with the dispersed gold nanoparticles. However, when a complementary DNA strand was added that brought

the particles closer together, the color turned purple within minutes. Since this initial work, an enormous effort has been focused on using nucleic acid based techniques to organize inorganic nanoparticles into periodic architectures. These structures have become important for fundamental studies focused on understanding the collective properties of nanoparticles^{90–92,109–114} and for the development of practical molecular diagnostic systems.^{8,9,115–119}

Much of our initial theory work in this area focused on disordered aggregates and superstructures;^{111,120} however, there was early interest in crystalline superlattices as well,^{121,122} which we have recently explored in great depth.^{104,123–125} These superstructures of particles can be considered metamaterials that are described by effective material parameters that dictate their properties. We also consider the mesoscale properties of these materials, where discrete superstructures can have complex optical mode interactions that can overshadow the plasmonic features underneath. Much of this work has required the application of light-scattering theories and solid-state material models that go beyond the simple theories described above.

1. Optical Properties of Three-Dimensional Plasmonic Arrays. Typically, the optical properties of 3D lattices of plasmonic nanoparticles appear similar to those of the free nanoparticle LSPR, though their extinction resonances are red-shifted due to changes in the background index, increased particle size, or decreased interparticle spacings. One way to describe these effects is in terms of effective bulk properties of the superlattice, with the key parameter being the effective dielectric function ϵ_{eff} which is dependent on metal volume fraction, the dielectric properties of the metal, and the background material.^{104,121–126} Maxwell–Garnett effective medium theory¹²⁷ (EMT) provides a means for the trivial calculation of ϵ_{eff} for a plasmonic superlattice that applies to the case where the nanoparticles are weakly coupled. Conceptually, an effective medium assumes that the properties of a material are the volume-weighted sum of its parts. Accurate up to ~20% metal volume, EMT provides a rapid means for the simulation of effective properties which can be used to inform and provide guidance for the *a priori* design of novel optical materials.^{104,121,123–126} Notably, defects and inhomogeneities in the lattice have minimal effect on the far-field optical response in low volume fraction systems; small quasistatic nanoparticles spaced on the order of their diameter are well described by EMT.¹²⁶ At close interparticle spacings (high volume fraction), or where the nanoparticles are very polydisperse such that a significant population is not quasistatic, it is necessary to use explicit particle calculations to accurately describe the optical response, as has sometimes been considered.^{122,123,126}

Within the effective medium formalism, the properties of the composite material are described as a volume-weighted sum of parts.¹²⁷ Formally, this can be written as

$$\frac{\epsilon_{\text{eff}} - \epsilon_m}{\epsilon_{\text{eff}} + 2\epsilon_m} = f \frac{\epsilon_i - \epsilon_m}{\epsilon_i + 2\epsilon_m} \quad (8)$$

where ϵ_i is the dielectric function of the “diluted material” inclusions; ϵ_m is the dielectric function of the background material; and f is the volume fraction of the inclusions. The DNA-programmed superlattices modeled here primarily consist of regularly spaced 20 nm diameter Au spheres separated by 20 base long DNA strands that are suspended in a background material (water) containing millimolar concentrations of sodium chloride (NaCl) and sodium phosphate buffer (Figure

5a, left). From EMT, one can generate a “bulk” dielectric function for the superlattice that is comprised of some

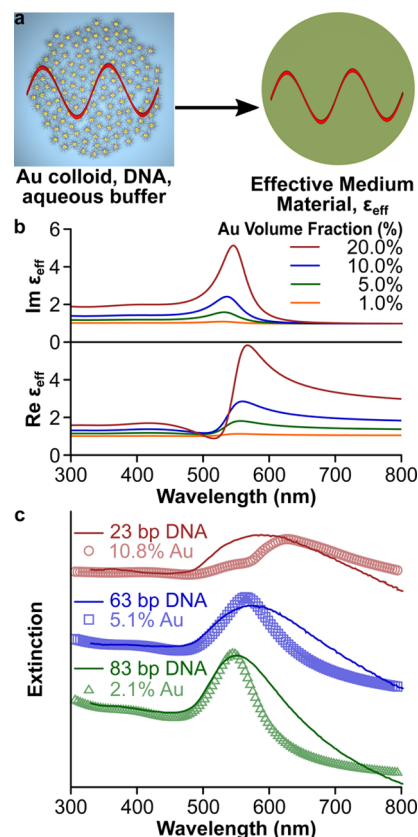


Figure 5. Effective material description of DNA–nanoparticle assemblies. (a) Schematic representation of a DNA–Au nanoparticle aggregate (left) as a homogeneous effective material (right). (b) Imaginary (top) and real (bottom) effective dielectric functions with varying Au volume fraction. (c) Extinction of DNA-mediated assemblies of 20 nm Au nanoparticles with varying DNA-linker lengths, i.e., varying number of base pairs, bp. Each base pair in a DNA duplex contributes ~0.255 nm.

percentage of the Au dielectric function (determined by volume fraction, typically 1–20% Au) and some fraction of the background dielectric (typically water or silica).^{97,124,125} This new “effective material” is now described by a homogeneous dielectric function, shown schematically in Figure 5a (right). ϵ_{eff} is easily related to several physical parameters that describe the optical properties of a material such as the effective refractive index and the penetration depth of light.¹²³

Several physical insights can be gathered from the complex effective dielectric function: $\text{Im}(\epsilon_{\text{eff}})$ is the imaginary dielectric function, a quantity which describes absorption and controls the spectrum of superlattices that are small compared to the wavelength. $\text{Re}(\epsilon_{\text{eff}})$ is the real part of the dielectric function, an important property that determines the ability of a material to concentrate electric charge (termed electric flux), and plays a key role in optical properties for mesoscale structures.^{123–125} Figure 5b plots ϵ_{eff} for a range of volume fractions of Au where EMT is accurate.¹⁰⁵ As the volume fraction increases (Figure 5b, top), the dominant peak increases in magnitude and red-shifts. For small particles, this captures the effect of weak dipole–dipole coupling between particles. This also agrees with

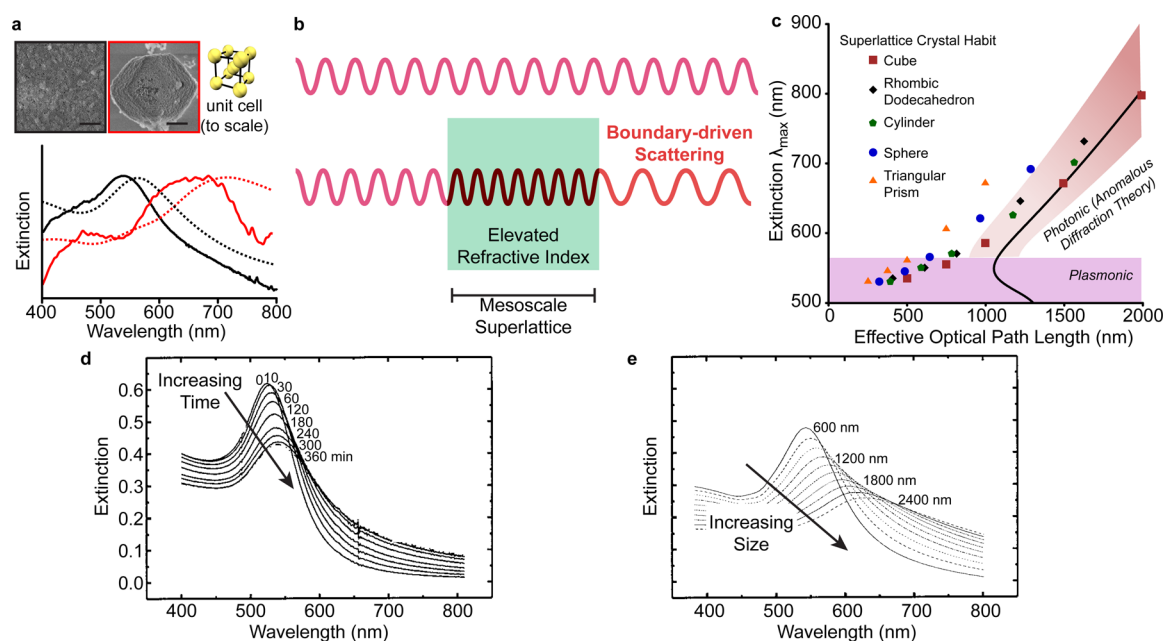


Figure 6. Size-dependent optical properties in DNA–nanoparticle superlattices. (a) Experimental (solid) and simulated (dashed) extinction spectra of a 150 nm thick thin film (black) and a $\sim 1 \mu\text{m}$ rhombic dodecahedron superlattice (red). Both structures have bcc unit cells (inset) of 20 nm diameter Au nanoparticles, ~ 15 nm surface-to-surface gap, $\sim 12\%$ Au, and simulations were performed using the coupled dipole approximation. (b) Schematic depiction of phase mismatch brought about by a mesoscale object with an elevated refractive index above background (green). (c) Size dependence for a variety of crystal habits (cube, rhombic dodecahedron, sphere, cylindrical disk, triangular prism) normalized by their size and volume. The black trace is the anomalous diffraction extinction approximation for 12.7% Au. (d) Extinction of amorphous DNA–Au nanoparticle aggregates over time (min) [72 base pair DNA, ~ 18 nm separation]. (e) Simulated extinction spectra of nanoparticle superlattices with a fixed volume fraction of 4.4% Au and increasing size (600–2400 nm diameter in 200 nm increments). Reprinted in part with permission from ref 111. Copyright 2001, American Chemical Society. Data adapted from ref 125.

experiment (Figure 5c), which depicts 20 nm radius Au spheres assembled into different volume fraction aggregates (i.e., different DNA linker lengths separating the particles). The extinction spectrum (which is primarily determined by absorption for small Au spheres) shows strong agreement with the simulation of a micron-sized sphere comprised of the effective dielectric function ϵ_{eff} . Discrepancies between the experimental and simulated data are largely due to size and shape effects of the aggregate in solution,^{104,111,125} which are challenging to isolate in amorphous systems (discussed in the next section).

In the years following the discovery of DNA-linked nanoparticle aggregates, much work was undertaken to determine what structural factors control the optical response. Early on, it was identified that particle size, aggregate size, and DNA length all contribute to the far-field optical properties of assemblies.^{90,111,112,121,122,128} Decreasing the DNA length, which increases the volume fraction of the aggregate, leads to red-shifting of the LSPR. In addition, it was determined that both increased nanoparticle size and increased aggregate size also can lead to red-shifting and dampening of the extinction maximum. However, the relationship between the aggregates' mesoscale structure and optical properties was challenging to elucidate. One issue is that the aggregates were polydisperse and fractal in nature, which greatly complicated modeling the effect of dipole–dipole interactions in the assembly.¹²⁰ Recently, it has been demonstrated that the far-field effects of changing the aggregate size and shape are nuanced, driven by complex interactions between the LSPRs of the constituent gold nanoparticles and photonic effects brought about by the size and shape of the aggregate.^{111,120,124,125} It was not until 3D

crystalline samples were synthesized that the relationship between nanoscale spacing and mesoscale architecture could be studied more thoroughly.

2. Size Effects in DNA-Assembled Three-Dimensional Plasmonic Superlattices. The transition from studying aggregates formed through an Ostwald ripening mechanism in solution to well-defined crystalline superlattices in the solid state has allowed for the elucidation of several explicit structure/property relationships. It has become apparent that superlattice size and shape can have a significant effect on the optical properties achievable with plasmonic nanoparticle superlattices. Early observations suggested that as the size of the aggregate increased the extinction red-shifted and dampened.^{111,121} Follow up studies demonstrated that the reaction-limited fractal structure of the superlattices explained much of the variation and broadening in extinction that occurred over time as the aggregates increased in size.¹²⁰ Additional theoretical studies that explored the effects of crystal symmetry and crystallite size for analogous systems demonstrated that both bcc and fcc crystals are readily described using EMT methods and that the far-field properties of large crystalline aggregates of arbitrary shape could be easily reproduced.^{111,121,122,128} However, rigorous characterization and spectroscopy of aggregates without defined shape remained a challenge.

In 2008, it was demonstrated that a nanoparticle densely functionalized with precisely designed DNA can function as a programmable atom equivalent (PAE),^{99,106,129} which can be crystallized into superlattices whereby the scale, symmetry, and composition can all be tuned independently.^{92,94–102,104–108} In recent years, control over the mesoscale structure of the

superlattices has been developed, such that single-crystal rhombic dodecahedron superlattices with defined crystal habits can be synthesized in addition to centimeter-scale thin films that are grown in a layer-by-layer fashion.^{101,102} Below, we discuss the optical properties of plasmonic assemblies primarily through the illustrative lens of DNA-programmed assembly, though many of the simulation frameworks and conclusions presented are general.

The recent discovery that large-area (centimeter-scale) thin films and single-crystal faceted rhombic dodecahedra could be reliably synthesized provides an ideal means for studying light–matter interactions in 3D plasmonic lattices in a highly controlled manner.^{101,102,124} For example, in 1 μm rhombic dodecahedron superlattices, it was identified that the extinction maximum could be red-shifted far (~ 150 nm) past the LSPR of the constituent gold nanoparticles.¹²⁵ Moreover, by changing the spacing between gold nanoparticles in the superlattice, this effect could be tuned such that at low volume fraction the extinction resembled that of the free nanoparticle LSPR. Strikingly, this did not occur in thin films (~ 150 nm thick), where the extinction maximum red-shifted ~ 15 nm with an equivalent decrease in interparticle spacing, commensurate with the red-shifts typically observed in solution (Figure 6a). These data suggest that the mesoscale size and shape of the superlattice play a pronounced role on the optical properties and that this effect can be controlled with great precision using DNA.

Importantly, the difference in extinction between large mesoscale superlattices and smaller thin films can be understood through EMT and geometric optics theories. First, recall that the real part of the dielectric function $\text{Re}(\epsilon_{\text{eff}})$ (Figure 5b) has a long elevated tail well to the red of the LSPR (reflecting preresonant dispersive behavior), such that the magnitude of $\text{Re}(\epsilon_{\text{eff}})$ is much larger than the background dielectric at wavelengths out into the infrared. Also, $\text{Im}(\epsilon_{\text{eff}})$ is close to zero so transmission losses are not important for micron size objects. As a result, the refractive index is almost entirely real. Its magnitude is elevated at these wavelengths, such that the exterior of the superlattice may have an index of $n = 1.33$ (water), while the interior of the superlattice could be as high as $n = 1.8$ – 2.0 , larger than most high-index polymers.^{130,131} For a large mesoscale object, this significant difference in refractive index will lead to phase mismatch (because light will travel at different speeds inside vs outside of the superlattice) (Figure 6b). In turn, this phase mismatch at the boundary of the superlattice leads to changes in the extinction intensity and location.¹²³ This scattering is well described by anomalous diffraction theory (ADT), a geometric optics scattering approximation commonly used to describe scattering by large water droplets and dust particles¹³² with the relation

$$\frac{2\pi l(n-1)}{\lambda} = n\pi \quad (9)$$

where l is the optical path length; n is the real refractive index; λ is the wavelength; and n (not to be confused with n) is the scattering order. Scattering maxima occur when n is an odd integer, corresponding to the phase mismatch condition. From eq 9, it is seen that the extent of phase mismatch can be controlled by either changing the interparticle spacing (which changes the real refractive index n , derived from $\text{Re}(\epsilon_{\text{eff}})$) or the superstructure size (changing the optical path length l , Figure 6c).

Importantly, these insights clarify data gathered in several previous studies,^{111,120,121} where both simulation and experiment suggest that increasing amorphous aggregate size results in increasingly red-shifted extinction maxima (Figure 6d,e). Only through the synthesis and stabilization of monodisperse and highly uniform crystalline lattices could these structure–property relationships be confirmed.^{123,125} These data also suggest that the far-field optical properties of the superlattice can be transitioned from plasmonic-dominated ($\text{Im}(\epsilon_{\text{eff}})$) to photonic-dominated through a combination of control over nanoparticle arrangement and crystal habit.^{111,123,125} Other recent work has identified that in large (~ 5 μm rhombic dodecahedra) the strength of Fabry–Perot resonances on the superlattice interior can be tuned by changing the interparticle spacing such that the plasmon either damps or enables oscillatory cavity modes depending on wavelength. Together, these works have important implications for optical devices in three-dimensional crystalline nanoparticle arrangements. A wide variety of crystal habits have been synthesized in the literature,^{102,133–140} several of which exhibited unique collective optical effects, suggesting that crystal habit could be a unique design parameter whose potential is unrealized in nanocrystalline systems.^{123,125,136,141} Using DNA as a means to dictate the crystal habit, in addition to the nanoparticle interparticle spacing and symmetry, provides an important framework for controlling the optical response of a plasmonic material *a priori*.

3. Emergent Properties and Unusual Optical Phenomena in Three-Dimensional Plasmonic Lattices. To this point, we have focused primarily on far-field extinction changes realized through superlattice morphology. However, by controlling the spacing and composition of the nanoscale components in a superlattice, the dielectric function can be tuned, and all aspects of the electromagnetic response can be manipulated with great precision. $\text{Re}(\epsilon_{\text{eff}})$ can be related to more complex behavior often observed in metamaterials, structures that have exquisitely designed optical properties that do not exist in nature.^{11,13,104,123} Recent works exploring these ideas have primarily involved the incorporation of multiple plasmonic materials within the lattice or using the shape of the particles to control the LSPR.^{104,123} The examples presented in this section provide new insight and understanding into the ability to control light propagation with the precise placement and control over nanoscale components.

For plasmonic materials, the use of Ag is often preferred to Au due to its narrower plasmon widths and greater tunability in LSPR location.²³ In the context of a plasmonic superlattice, it is expected that the incorporation of Ag particles will allow for more dispersive resonances, greater spectral tunability, and tailorable metamaterial response. With the use of Ag in combination with anisotropic nanoparticles, unique spectral behavior can be achieved.¹²³ Using a variety of theoretical methods, Ross et al. demonstrated that a variety of exotic optical properties are achievable with any high aspect ratio Ag plasmonic nanoparticle at surprisingly low volume fraction.¹²³ In this work, ellipsoids were used as a proxy for high aspect ratio nanoparticles, such as rods, triangular prisms, or cylindrical disks, all of which can be synthesized using wet chemical methods and assembled with great fidelity using a variety of techniques.³⁹ It was demonstrated that at volume fractions as low as 1.0% Ag the real dielectric function $\text{Re}(\epsilon_{\text{eff}})$ becomes negative (recall that for Au, Figure 5b, $\text{Re}(\epsilon_{\text{eff}})$ remained positive for all wavelengths). In experimental systems of closely packed Ag nanospheres, ellipsometry measurements

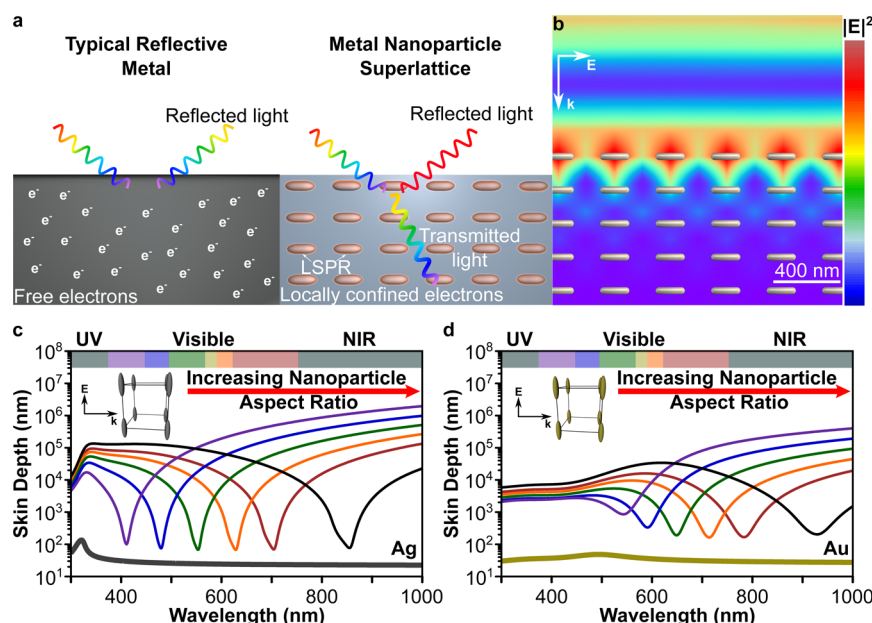


Figure 7. Emergent optical properties of plasmonic nanoparticle superlattices. (a) Scheme of reflection from a typical metal (left) and from a metal nanoparticle superlattice (right). (b) Electric field profile at the skin depth minimum for a superlattice comprised of 80 nm long $\times$ 10 nm radius ellipsoids. Skin penetration depth for Ag (c) and Au (d) superlattices of ellipsoids with increasing aspect ratio compared to the bulk metal skin depth. The electric field is plotted on a logarithmic scale from 0.01 to 10. Data adapted from ref 123.

of the effective dielectric function have confirmed that this negative behavior can exist within a finite spectral window.¹³¹ Near-zero and negative $\text{Re}(\epsilon_{\text{eff}})$ are noteworthy because (1) at $\text{Re}(\epsilon_{\text{eff}}) \sim 0$ the wavelength of light in a material is effectively infinite and light can “tunnel” across a boundary of arbitrary size or shape^{142,143} and (2) at negative values of $\text{Re}(\epsilon_{\text{eff}})$, a material will act metallic, meaning that the material is resistant to supporting electromagnetic waves and will act glossy and reflective like a mirror. By changing the aspect ratio of the anisotropic particles, this behavior can be achieved throughout the visible and near-infrared.¹²³

A simple measure of comparison for optical behavior in plasmonic superlattices is the skin depth, a metric that controls the extent to which light penetrates a given material. For bulk noble metals, the skin depth is between 20 and 30 nm, whereas the values for common dielectrics such as water (10^4 nm) and glass (10^8 nm) are much larger. In a metal, the skin depth is low primarily because incident light on a metal is rapidly absorbed and scattered by electrons (Figure 7a). A plasmonic superlattice, however, is not conductive because the electrons are confined to the individual nanoparticles. Thus, the superlattice acts as a diluted metal, whereby wavelengths of light near the LSPR will interact as if the material is metallic and other wavelengths will transmit as if the material were dielectric (Figure 7b). The skin depth functions for lattices of anisotropic ellipsoids with varying aspect ratios are depicted in Figure 7c,d (Ag, Au). Strikingly, these data suggest that a material that is less than 1% Ag (and thus 99% air or water) attenuates light as efficiently as a bulk conductive metal at some wavelengths. This demonstration suggests that many remarkable properties are achievable with lattices of anisotropic plasmonic nanostructures, a major focus of promising optical properties and challenging new assemblies in this field. Similar unusual dielectric behaviors can also be achieved with Ag spheres, albeit at larger volume fractions (~ 15 – 20%) and shorter wavelengths.¹⁰⁴ Notably, near-field coupled plasmonic arrays

with much higher volume fractions can create metal–insulator transitions and unique reflective properties,^{144–147} in addition to tunable metallic dielectric functions;^{148–150} several of these systems are described in detail in Supporting Information, Section S.2.

Finally, DNA-programmable crystallization provides an important means for studying interactions between different plasmonic materials. CsCl-like binary lattices of Ag and Au nanospheres demonstrated significant dampening and broadening of the Ag LSPR with decreasing interparticle spacing.¹⁰⁴ This effect was attributed to a Fano-like interference between the Ag sphere LSPR and the Au sphere interband transitions (inherent absorption due to transitions in the d- and f-electron states) that could be controlled by changing the interparticle spacing. A similar effect was first observed in dimers assembled with DNA, which was explained by the plasmon-hybridization model.¹⁵¹ This provides an important demonstration that precise control using DNA can be used to finely tune the interactions between plasmonic nanoparticles with different composition, a relatively unexplored means for achieving unique optical materials.

VI. FUTURE OUTLOOK

A common theme for all of the works described above is that the precise manipulation and orientation of plasmonic nanoparticles across multiple length scales can lead to diverse and exciting optical properties. Rapid advances in theoretical methods have provided target structures for plasmonic materials, resulting in a variety of proof-of-concept devices and functions.^{80,124,152–155} These new assemblies continue to challenge the theories that we use to describe plasmonic materials. Describing systems comprised of spheres and ellipsoids (which can be a proxy for high-aspect ratio nanoparticles) has proven successful;^{3,68,72,123,128} however an immense array of nanoparticle shapes exist that can be arranged into complex symmetries and orientations. Understanding the

properties of these systems built from complex anisotropic building blocks will require new methods for extrapolating complex near-field coupling into far-field bulk properties.^{40,51,156–159} Finally, the union of photonic and plasmonic effects in three-dimensions is an early and promising means for the continued development of visible-light metamaterials, creation of optoelectronic plasmonic devices and sensors, and the use of simple theoretical methods to describe complex media with deliberately designed properties.^{11,13,63,80,95,123–125,160} The ability to leverage the exquisite sensitivity and collective behavior of plasmonic nanoparticles is crucial to understanding chemical and physical phenomena on the nanoscale. Thus, we conclude that one-, two-, and three-dimensional plasmonic assemblies have a rich outlook.

■ ASSOCIATED CONTENT

● Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.jpcc.5b10800.

Additional figures, equations, and the detailed description of retardation effects in two-dimensional plasmonic arrays are included in the Supporting Information in addition to a literature survey of near-field coupled plasmonic arrays(PDF)

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The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.

Notes

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